

AI Feature UI Integration Plan

National Land Acquisition & Management System — SIH 2026

How to wire Document Mismatch Detection and Risk Scoring into the existing frontend

This document describes where and how the two AI features should appear in the existing application UI, based on the current design already in place (the Parcels page, sidebar navigation, and status badge patterns). It assumes the AI Features Reference document (API fields, example responses) as a companion — this document focuses on placement and UX, not the raw API contract.

Summary of Placement

Feature	Where it lives in the UI
Document Mismatch Detection	Existing “AI Mismatch” sidebar page — expanded into a full list + detail view, plus a flag indicator on the Parcels page.
Risk Scoring	No new sidebar item needed — lives on the Project detail page (score badge + breakdown) and on the Dashboard's High-Risk Projects KPI.

Feature 1 — Document Mismatch Detection

The sidebar already has an “AI Mismatch” entry with no content built yet. This becomes the home for the feature, structured as three views.

1. Mismatch List (main view)

A table in the same visual style as the existing Parcels page — one row per detected mismatch, not per document, so an officer can scan for issues quickly.

Columns

- Parcel Code (e.g. P-101) — links back to the parcel's record.
- Field — which field disagreed (Area, Village, Survey No, Owner Name, District).
- Official Value vs Extracted Value — shown side-by-side.
- Severity — colored badge, same visual pattern as the existing Acquisition Status badges (e.g. NOTIFIED, ACQUIRED).
- Status — DETECTED / UNDER_REVIEW / RESOLVED / FALSE_POSITIVE.
- Detected Date.
- Action — “View Record” button, matching the existing table's action column.

Filters (top of page, same style as the Parcels search bar)

- Search by parcel code, survey number, or village.
- Filter by Project (dropdown, “All Projects” default — identical pattern to the Parcels page).
- Filter by Severity (LOW / MEDIUM / HIGH / CRITICAL).

- Filter by Status (“All Statuses” default, same pattern as Parcels page).

2. Mismatch Detail (clicking a row)

Opens a detail panel or page showing:

- A side-by-side comparison card: Official Value vs Extracted Value for the flagged field.
- The plain-language explanation sentence returned by the API — shown as-is, no further processing needed.
- A severity badge, matching the list view.
- An officer action: a status dropdown (Under Review / Resolved / False Positive) so a human makes the final call — this keeps the feature framed as decision support, not an automatic verdict.
- A link back to the source document and the parcel record.

3. “Run New Check” Action

A button in the same position/style as the existing “+ Add Land Parcel” button (top-right of the page header).

Opens a small form:

- Select a document (dropdown or upload).
- Select the parcel to check it against.
- Submit → calls the compare endpoint → result appears at the top of the Mismatch List.

Connecting back to the Parcels page

Add a small flag icon or badge next to any parcel row that has unresolved mismatches — for example, a red flag icon next to P-101 if it currently has an open mismatch. This uses data you already have (mismatches linked to `parcel_id`) and creates a natural link between the two pages without duplicating any UI.

Feature 2 — Risk Scoring

Risk scoring is calculated at the project level, not the parcel level, so it doesn't need its own sidebar entry — it fits naturally into two places that already exist in the information architecture.

1. Project Detail Page

- A prominent score + risk-level badge near the top of the page — reuse the existing badge visual style (rounded pill, colored background) already used for Acquisition Status.
- Suggested color mapping, consistent with the badge style already visible in the app (green/blue/amber/red family):
 - LOW → green (same family as “ACQUIRED” / “POSSESSION TAKEN”)
 - MEDIUM → amber (same family as “NOTIFIED”)
 - HIGH / CRITICAL → red (same family as “R&R ISSUE / FLAGGED”)
- An expandable “Why this score?” section showing the four contributing factors as labeled progress bars (points contributed out of each factor's maximum) — this is the feature's key strength over a black-box model, so it deserves visual prominence, not a buried tooltip.

2. Dashboard — High-Risk Projects

The system's dashboard structure already defines a “High-Risk Projects” KPI card. This becomes a real, functioning widget:

- A count or list of projects currently at HIGH or CRITICAL risk_level.
- Clicking through leads to the relevant Project Detail page, where the full breakdown is visible.
- Since risk scores are stored historically (not overwritten on recalculation), a small score-over-time sparkline is possible here later without any backend changes.

Suggested Build Order

1. AI Mismatch list page — highest visibility, sidebar link already exists.
2. Mismatch detail view + officer status update.
3. “Run New Check” trigger form.
4. Flag indicator on the existing Parcels page.
5. Risk score badge + breakdown on the Project detail page.
6. Dashboard High-Risk Projects widget.

Design Principles to Carry Through the UI

- Both features are decision support, not automated verdicts — always give the officer a way to review and act, never present a result as final.
- Reuse existing visual patterns (badges, table layout, filter bar, button placement) rather than introducing new UI conventions — the screenshots already establish a clean, consistent style worth matching exactly.
- The risk score's factor breakdown is the feature's main selling point — don't hide it behind an extra click if avoidable; showing the “why” is what separates this from an opaque score.